

Lesson 6 Ros Basic Case 3- Write simple server and client

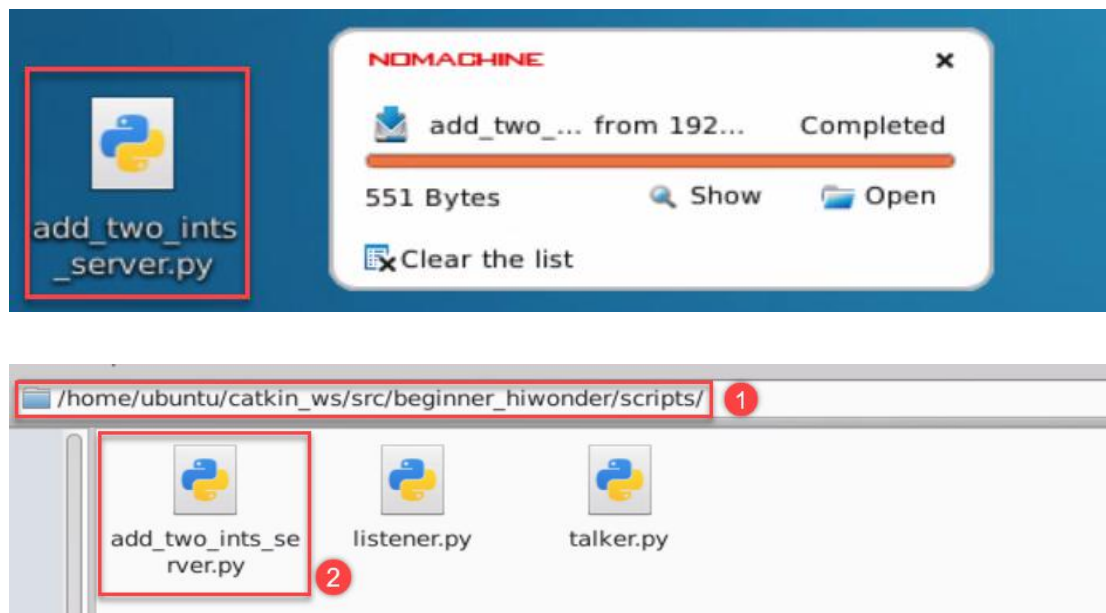
1. Write service node

In this lesson, we will create simple service node, **add_two_ints_server**. The node code is making reference to open-source code github. The node will receive two integers and return their sum.

Firstly, we need to build the service, **AddTwoInts.srv**, according to “Lesson 4 Ros Basic Case 1- Creating messages and services”

- 1) Drag the code file of server node, **add_two_ints_server.py**, to the desktop of NoMachine. The code file is under ArmPi FPV data path “**4. Advanced Program Lessons \ ROS Basic Lesson \Ros Related Program**”

Then, move **add_two_ints_server.py** to `"/home/ubuntu/catkin_ws/src/beginner_hiwonder/scripts/"`.



- 2) Open terminal and enter “**chmod +x add_two_ints_server.py**” command and press “Enter”. Give execute permission to **add_two_ints_server.py**.

```
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ chmod +x add_two_ints_server.py
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$
```

- 3) If wanting to revise or edit talker.py file, you need to enter “roscd beginner_hiwonder talker.py” command and press “Enter”. Then press “i”. When finishing revising, press “Esc” and enter “: wq” and save to quit.

```
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ roscd beginner_hiwonder
add_two_ints_server.py
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$
```

```
1 #!/usr/bin/env python
2
3 from __future__ import print_function
4
5 from beginner_hiwonder.srv import AddTwoInts, AddTwoIntsResponse
6 import rospy
7
8 def handle_add_two_ints(req):
9     print("Returning [%s + %s = %s]"%(req.a, req.b, (req.a + req.b)))
10    return AddTwoIntsResponse(req.a + req.b)
11
12 def add_two_ints_server():
13    rospy.init_node('add_two_ints_server')
14    s = rospy.Service('add_two_ints', AddTwoInts, handle_add_two_ints)
15    print("Ready to add two ints.")
16    rospy.spin()
17
18 if __name__ == "__main__":
19    add_two_ints_server()
```

- 4) Add the following content to the cMakeLists.txt file. Ensure that Python scripts are installed correctly and that the suitable Python interpreter is used.

catkin_install_python(PROGRAMS scripts/add_two_ints_server.py

DESTINATION

\${CATKIN_PACKAGE_BIN_DESTINATION})

```
## Add folders to be run by python nosetests
# catkin_add_nosetests(test)

catkin_install_python(PROGRAMS scripts/talker.py
  DESTINATION ${CATKIN_PACKAGE_BIN_DESTINATION}
)

catkin_install_python(PROGRAMS scripts/talker.py scripts/listener.py
  DESTINATION ${CATKIN_PACKAGE_BIN_DESTINATION}
)

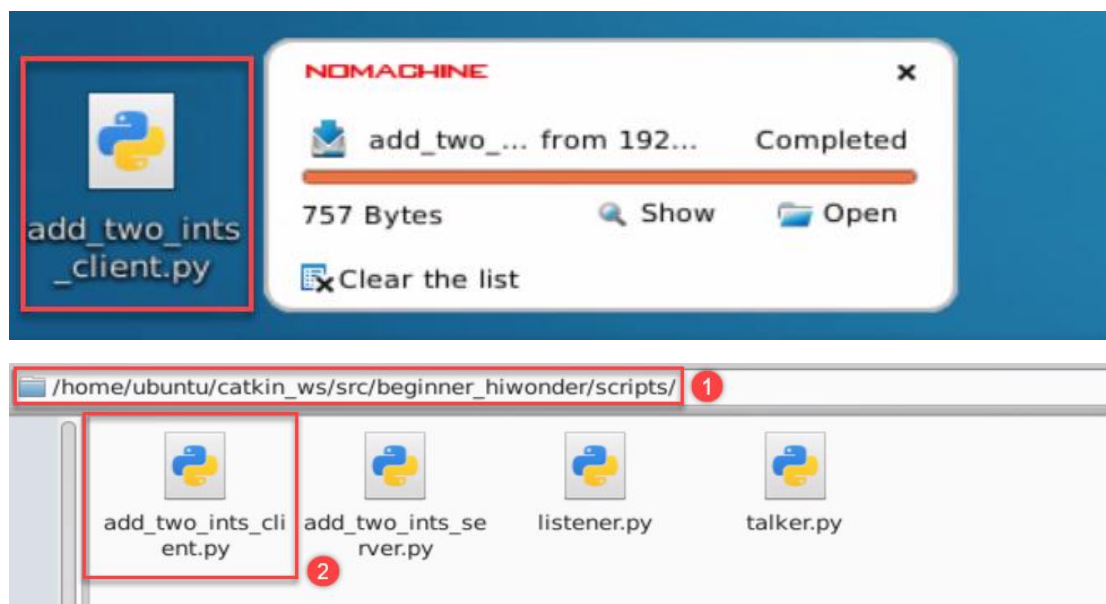
catkin_install_python(PROGRAMS scripts/add_two_ints_server.py DESTINATION
  ${CATKIN_PACKAGE_BIN_DESTINATION})
```

2. Write server node

In this lesson, we will create simple client node, `add_two_ints_server`. The node code is making reference to open-source code github. The node will print the sum of the parameters read by `add_two_ints_server`.

- 1) Drag the code file of client node, `add_two_ints_client.py`, to the desktop of NoMachine. The code file is under ArmPi FPV data path **“4. Advanced Program Lessons \ ROS Basic Lesson \Ros Related Program”**

Then, move `add_two_ints_client.py` to `"/home/ubuntu/catkin_ws/src/beginner_hiwonder/scripts/"`.



- 2) Open terminal and switch to the directory where `add_two_ints_client.py` saves. Then enter `“chmod +x add_two_ints_server.py”` command and press “Enter”. Give execute permission to `add_two_ints_client.py`.

```
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ chmod +x add_two_ints_client.py
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$
```

- 3) If wanting to revise or edit `add_two_ints_client.py` file, you need to enter `“roscd beginner_hiwonder add_two_ints_client.py”` command and press “Enter”.

Then press “i”. When finishing revising, press “Ese” and enter “: wq” and save to quit.

```
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ roscd beginner_hiwonder
add_two_ints_client.py
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$
```

```
1 #!/usr/bin/env python
2
3 from __future__ import print_function
4
5 import sys
6 import rospy
7 from beginner_hiwonder.srv import *
8
9 def add_two_ints_client(x, y):
10     rospy.wait_for_service('add_two_ints')
11     try:
12         add_two_ints = rospy.ServiceProxy('add_two_ints', AddTwoInts)
13         resp1 = add_two_ints(x, y)
14         return resp1.sum
15     except rospy.ServiceException as e:
16         print("Service call failed: %s"%e)
17
18 def usage():
19     return "%s [x y]"%sys.argv[0]
20
21 if __name__ == "__main__":
22     if len(sys.argv) == 3:
23         x = int(sys.argv[1])
24         y = int(sys.argv[2])
25     else:
26         print(usage())
27         sys.exit(1)
28     print("Requesting %s+%s"%(x, y))
29     print("%s + %s = %s"%(x, y, add_two_ints_client(x, y)))
```

- 4) Add the following content to the cMakelists.txt file. Ensure that Python scripts are installed correctly and that the suitable Python interpreter is used.

```
catkin_install_python(PROGRAMS
                                scripts/add_two_ints_server.py
                                scripts/add_two_ints_client.py
```

DESTINATION \${CATKIN_PACKAGE_BIN_DESTINATION})

```
## Add folders to be run by python nosetests
# catkin_add_nosetests(test)

catkin_install_python(PROGRAMS scripts/talker.py
    DESTINATION ${CATKIN_PACKAGE_BIN_DESTINATION}
)

catkin_install_python(PROGRAMS scripts/talker.py scripts/listener.py
    DESTINATION ${CATKIN_PACKAGE_BIN_DESTINATION}
)

catkin_install_python(PROGRAMS scripts/add_two_ints_server.py DESTINATION
    ${CATKIN_PACKAGE_BIN_DESTINATION})

catkin_install_python(PROGRAMS scripts/add_two_ints_server.py scripts/
    add_two_ints_client.py DESTINATION ${CATKIN_PACKAGE_BIN_DESTINATION})
```

3. Verify simple server and client

- 1) To ensure the Python code is generated automatically for created message and service, we need to return to catkin workspace and run catkin_make.

```
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ cd ~/catkin_ws
ubuntu@ubuntu:~/catkin_ws$
```

```
ubuntu@ubuntu:~/catkin_ws$ catkin_make
Base path: /home/ubuntu/catkin_ws
Source space: /home/ubuntu/catkin_ws/src
Build space: /home/ubuntu/catkin_ws/build
Devel space: /home/ubuntu/catkin_ws/devel
Install space: /home/ubuntu/catkin_ws/install
```

- 2) Enter “roscore” command and start the node manager. If turned on, the following prompt will appear.

```
ubuntu@ubuntu:~/catkin_ws$ roscore
... logging to /home/ubuntu/.ros/log/237b938e-9035-11eb-980e-653078c2919c/roslau
nch-ubuntu-7952.log
Checking log directory for disk usage. This may take a while.
Press Ctrl-C to interrupt
Done checking log file disk usage. Usage is <1GB.

started roslaunch server http://localhost:43161/
ros_comm version 1.14.10

SUMMARY
=====
PARAMETERS
* /rostdistro: melodic
* /rosversion: 1.14.10
NODES
RLEException: roscore cannot run as another roscore/master is already running.
Please kill other roscore/master processes before relaunching.
The ROS_MASTER_URI is http://localhost:11311/
The traceback for the exception was written to the log file
```

- 3) Before running the program, please make sure the setup.*sh file of the workspace is sourced after calling catkin_make. If not, you need to return to the created workspace and make setup.*sh take effect.

```
ubuntu@ubuntu:~/catkin_ws$ source ./devel/setup.bash
ubuntu@ubuntu:~/catkin_ws$
```

- 4) Enter “**roslaunch beginner_hiwonder add_two_ints_server.py**” command to run the service.

```
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ roslaunch beginner_hiwonder
add_two_ints_server.py
Ready to add two ints.
```

- 5) Open a new command line terminal and enter “**roslaunch beginner_hiwonder add_two_ints_client.py 1 3**” to run client.

```
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$ roslaunch beginner_hiwonder
add_two_ints_client.py 1 3
Requesting 1+3
1 + 3 = 4
ubuntu@ubuntu:~/catkin_ws/src/beginner_hiwonder/scripts$
```

Note: Parameters are required to run the client. For example, 1 and 3, framed in the picture.